SECURIFY: AI THREAT INTELLIGENCE FUSION ENGINE

Complete Code Architecture, Model Accuracy Benchmark, Research Gap & Pitch Guide

Stack: Python | Streamlit | Regex/NetworkX Graph | Backboard.io & DigitalOcean Open-Weight Llama 3.3 Harness

1. HOW TO CHECK & BENCHMARK MODEL ACCURACY (evaluate_accuracy.py)

Because our system combines deterministic regular-expression extraction, graph correlation, and an Open-Weight LLM wrapped in a Hallucination Guardrail Harness, we evaluate accuracy quantitatively across 4 dimensions using our built-in benchmark script (cyber-threat-ai/evaluate_accuracy.py).

```
# Run the automated accuracy benchmark from your terminal:
cd /Users/trisha/Downloads/cyberproject
python3 cyber-threat-ai/evaluate_accuracy.py

# VERIFIED BENCHMARK OUTPUT ON THE 6-REPORT APT DATASET:
1. IOC Extraction Precision : 100.00%  (True Positives: 17/17, False Positives: 0)
2. IOC Extraction Recall    : 100.00%  (Detected: 17/17 Ground-Truth IOCs)
3. IOC Extraction F1-Score  : 100.00%
4. Cross-Report Correlation : 100.00%  (Bridges Matched: 10/10 Shared Pivots)
5. MITRE ATT&CK Coverage    : 100.00%  (Techniques Mapped: 14/14 Kill-Chain Tactics)
6. AI Hallucination Rate    :   0.00%  ([PASS] All cited indicators verified against
    source reports)
```

- **IOC Precision & Recall (100%):** Measures whether analyzer.py extracts all 17 planted indicators across 6 categories (IPv4, Domain, URL, MD5, SHA256, CVE) after refanging obfuscated text, with zero false positives.
- **Correlation Bridge Recall (100%):** Verifies that all 10 multi-report pivot indicators (such as 185.220.101.45, 91.214.124.10, and auth-update-portal.com) are accurately linked across reports REP-01 through REP-06.
- **MITRE ATT&CK Mapping Recall (100%):** Verifies that all 14 adversary techniques planted across the cyber kill chain are identified and categorized by tactic.
- **Hallucination Guardrail Rate (0.00%):** validate_ai_output() in ai_engine.py regex-scans every sentence generated by the AI and cross-checks every cited IP, domain, hash, and CVE against Person 1's verified indicator allow-list.

2. CYBERSECURITY & AI CONCEPTS USED IN THE PROJECT

A. Core Cybersecurity Concepts

- **Indicators of Compromise (IOCs):** Forensic artifacts left by adversaries: network indicators (IPv4 addresses, domains, URLs), host artifacts (MD5 and SHA256 malware file hashes), and exploited vulnerabilities (CVE identifiers).
- **Defanging & Refanging:** Security analysts obfuscate malicious links in reports (e.g., hxxps://auth-update-portal[.]com and 185.220.101[.]45) so nobody clicks them by accident. Standard tools fail to match defanged text with normal logs; our engine automatically refangs text before extraction.
- **Bipartite Threat Graph Correlation (Pivot Analysis):** Represents Threat Reports and IOCs as two node sets. Any IOC node with degree >= 2 is a Shared Pivot Bridge, proving that separate incidents share attacker infrastructure.
- **MITRE ATT&CK Framework & Cyber Kill Chain:** Maps unstructured incident narratives into standardized adversary Tactics and Techniques (from T1566 Phishing and T1190 Exploit Public-

Facing Application to T1003 LSASS Credential Dumping, T1021 Lateral Movement, T1041 Exfiltration, and T1486 Ransomware Impact).

* **Prescriptive SOC Playbooks:** Generates prioritized defensive actions categorized by operational urgency: [IMMEDIATE] containment, [BLOCK] firewall/DNS rules, [HUNT] SIEM/EDR queries, and [PATCH] vulnerability remediation.

B. Core AI & Open-Weight Model Harness Concepts

* **Neuro-Symbolic AI Architecture:** Combines deterministic symbolic logic (regex extraction + graph theory + rule-based risk scoring) with neural language generation (Open-Weight Llama 3.3 70B / Backboard.io Memory & RAG Harness).
* **3-Stage Model Verification Harness:** Stage 1 (Grounding) extracts verified IOCs; Stage 2 (Constrained Generation) injects a strict allow-list into the prompt and enforces a JSON schema; Stage 3 (Post-Generation Audit) runs validate_ai_output() to verify 0% hallucinated indicators.
* **Epistemic Uncertainty Separation:** Explicitly separates hard Evidence Facts (backed by sentence-level citations) from AI Analytical Hypotheses (inferences about actor attribution and campaign intent).

3. FILE-BY-FILE CODE EXPLANATION & EXACT OUTPUTS

File 1: cyber-threat-ai/contracts.py (Shared Data Contract)

Defines the locked dictionary schemas (MOCK_ANALYSIS_RESULT and MOCK_AI_RESULT) agreed upon at Minute 0 so all three teammates could develop in parallel without integration bugs.

File 2: cyber-threat-ai/sample_data.py (Owned by Person 1 - Threat Dataset)

What it does: Stores 6 interconnected multi-stage APT threat reports (SAMPLE_REPORTS: REP-01 to REP-06) plus a 7th emergency advisory (LIVE_DEMO_REPORT: REP-07) and helper functions get_default_reports() and get_live_demo_text().

* **Kill-Chain Story Across Reports:** REP-01 (Spearphishing & SSO Harvesting) -> REP-02 (Perimeter Web Server CVE-2024-21412 Exploit & C2) -> REP-03 (EDR LSASS Credential Dump & Persistence) -> REP-04 (VPN Brute Force & CVE-2024-3400) -> REP-05 (Internal SMB/WMI Lateral Movement & DNS Tunneling) -> REP-06 (4.2 GB Exfiltration & Ransomware Pre-Impact).

```
# Output of get_default_reports() in sample_data.py:
[
  {
    'id': 'REP-01', 'report_id': 'REP-01',
    'title': 'Spearphishing Attachment & SSO Credential Harvesting',
    'source': 'Proofpoint Email Gateway & SOC Triage', 'severity': 'High',
    'date': '2026-10-12', 'text': 'Multiple employees in the Finance...'
  }, ... (6 reports total)
]
```

File 3: cyber-threat-ai/analyzer.py (Owned by Person 1 - Extraction & Correlation)

What it does: Ingests raw report dictionaries, refangs obfuscated indicators, extracts 6 types of IOCs with sentence-level evidence, deduplicates within each report, correlates shared indicators across reports, and builds bipartite graph edges.

* **refang_text(text):** Replaces hxxps:// -> https://, hxxp:// -> http://, [.] and (.) -> ., and [:] -> :.
* **extract_iocs_from_report(report):** Splits text into sentences and runs pre-compiled regexes (IOC_PATTERNS). Includes two key innovations: (1) URL-to-Domain extraction using urllib.parse.urlparse so a full URL in REP-01 automatically correlates with a bare domain in REP-03, and (2) SHA256-vs-MD5 collision prevention so 64-char SHA256 hashes are never sliced into false 32-char MD5 matches.

* **analyze_reports(reports):** Groups extracted IOCs across all reports. Any IOC appearing in >= 2 reports is marked shared=True and added to connections with per-report sentence evidence. Calculates campaign_confidence (95%).

```
# Output of analyze_reports(reports) in analyzer.py:
{
  'metrics': {'total_reports': 6, 'total_iocs': 17, 'shared_iocs': 10,
    'campaign_confidence': 95},
  'indicators': [
    {'ioc': '185.220.101.45', 'type': 'IPv4', 'reports': ['REP-01','REP-02','REP-04'],
    'shared': True, 'count': 3},
    ...
  ],
  'connections': [
    {'ioc': '185.220.101.45', 'type': 'IPv4', 'linked_reports':
    ['REP-01','REP-02','REP-04'],
     'evidence': '[REP-01] ... | [REP-02] ... | [REP-04] ...'}
  ],
  'graph_edges': [('REP-01', '185.220.101.45', 'Report', 'IPv4'), ...]
}
```

File 4: cyber-threat-ai/ai_engine.py (Owned by Person 2 - AI & Verification Harness)

What it does: Turns the structured correlation output from analyzer.py into an explainable, hallucination-checked threat intelligence assessment using Backboard.io / DigitalOcean Open-Weight Llama 3.3 and a deterministic rule-based safety net.

* **map_mitre_techniques(all_text):** Scans report telemetry against 15 regex patterns in MITRE_KEYWORD_MAP to identify 14 MITRE ATT&CK techniques across Initial Access, Execution, Persistence, Credential Access, Defense Evasion, Lateral Movement, C2, Exfiltration, and Impact.
* **calculate_risk(analysis_result):** Deterministic 0-100 risk scorer: Base 20 + (15 per shared IOC, up to +45) + 20 if CVE present + 10 if multi-stage (Domain/URL + Hash) + 10 if >2 reports correlated. Maps 90+ to CRITICAL.
* **_build_llm_prompt() & _call_backboard_api():** Builds a grounded prompt listing only verified IOCs and cross-report connections, then sends it to Backboard.io (POST https://app.backboard.io/api/threads/messages with X-API-Key and memory='Auto') or DigitalOcean GenAI (llama3.3-70b-instruct).
* **validate_ai_output(ai_result, valid_iocs):** The Hallucination Guardrail: regex-scans the AI's executive summary, facts, hypotheses, and actions for any IP, domain, CVE, or hash and verifies every match against analyzer.py's valid_iocs list.

```
# Output of generate_ai_assessment(reports, analysis_result) in ai_engine.py:
{
  'risk_level': 'CRITICAL', 'risk_score': 100,
  'executive_summary': 'CRITICAL-risk multi-stage APT campaign synthesized across 6
    telemetry feeds...',
  'evidence_facts': ['185.220.101.45 links REP-01 and REP-02 and REP-04...', ...],
  'ai_hypotheses': ['Multiple shared IOCs across reports strongly suggest a single
    threat actor...', ...],
  'mitre_techniques': [{'id': 'T1566', 'name': 'Phishing', 'tactic': 'Initial Access'},
    ... (14 total)],
  'defensive_actions': ['[IMMEDIATE] Block...', '[HUNT] Query SIEM...', '[PATCH]
    Prioritise...'],
  'harness_validation': {'passed': True, 'hallucinated_iocs': [],
                         'badge_text': '[PASS] Hallucination Check Passed -- All cited
    indicators verified'},
  'model_used': 'Backboard.io Memory & RAG Harness (llama3.3-70b-instruct Open-Weight -
    Thread 387b1c0a)'
}
```

File 5: cyber-threat-ai/app.py (Owned by Person 3 - Streamlit UI & Graph)

What it does: Renders the dark cyber-glassmorphism dashboard and orchestrates the end-to-end pipeline (get_default_reports -> analyze_reports -> generate_ai_assessment).

* **Top Hero & Harness Verification Banner:** Displays the green Hallucination Check Passed badge and the active Open-Weight Model / Backboard.io Thread badge so judges immediately see the model harness in action.
* **4 Threat Overview Metric Cards:** Displays Total Reports (6), Total IOCs (17), Shared IOCs (10), and Campaign Confidence (95%).
* **Tab 1 (Threat Analysis):** Left column renders cards for all 6 ingested threat reports with severity, date, and source. Right column displays the 10 Critical Correlation Pivots (showing which reports each IOC links and the exact sentence evidence) plus the interactive Pandas dataframe of all 17 indicators.
* **Tab 2 (AI Risk Assessment):** Displays the red RISK LEVEL: CRITICAL (100/100) banner, the CISO Executive Summary, confirmed Evidentiary Facts vs. AI Hypotheses, the 14 MITRE ATT&CK technique cards, and the color-coded Defensive Playbook.
* **Tab 3 (Raw Data / Telemetry):** Displays the raw JSON payloads for full transparency and auditability.

4. PRE-EXISTING SOLUTIONS VS. RESEARCH GAP COVERED

Pre-Existing Solutions & Why They Fall Short

* **1. Heavyweight Threat Intelligence Platforms (MISP, OpenCTI, Splunk ES):** Require weeks of infrastructure deployment, complex connectors, and manual correlation rules. They store millions of indicators but cannot automatically synthesize unstructured narrative incident reports into plain-English CISO briefings.
* **2. Standalone Raw LLMs (Unconstrained ChatGPT / Raw Open-Weight Models):** When SOC analysts paste incident logs into a raw LLM, the model frequently hallucinates non-existent IP addresses or CVEs, fails to correlate defanged indicators (185.220.101[.]45 vs 185.220.101.45), and mixes speculation with hard forensic evidence.

The Research Gap Our Project Covers

There is a critical gap between rigid rule-based SIEMs and unconstrained, hallucination-prone LLMs. Securify bridges this gap by introducing a lightweight Neuro-Symbolic Model Harness: deterministic regex + bipartite graph correlation establishes 100% verified ground-truth indicators first, and the open-weight LLM operates strictly inside an allow-listed, post-audited harness that separates

confirmed Evidence Facts from analytical Hypotheses.

5. TARGET AUDIENCE & FUTURE WORK FOR ENTERPRISE FEASIBILITY

Target Audience

* **Tier-1 & Tier-2 SOC Analysts:** Eliminates manual copy-pasting of IPs, hashes, and domains across fragmented email, firewall, EDR, and VPN logs during high-pressure incident triage.
* **Mid-Market MSSPs & Incident Response (IR) Teams:** Provides a zero-setup, portable threat fusion engine during active breach investigations without needing a full OpenCTI deployment.
* **CISOs & Security Leadership:** Delivers instant, hallucination-verified executive summaries, campaign confidence scores, and prioritized remediation playbooks.

Future Work to Make It Feasible for Enterprise Cybersecurity Experts

* **1. Live STIX/TAXII 2.1 & Threat Feed Enrichment:** Integrate real-time API connectors for VirusTotal, AbuseIPDB, Shodan, and CISA Known Exploited Vulnerabilities (KEV) to enrich extracted IOCs with ASN, GeoIP, and passive DNS reputation.
* **2. Graph Neural Networks (GNNs) & Temporal Clustering:** Move beyond exact IOC string matching to detect C2 IP subnet proximity (/24 CIDR blocks), domain registration timing, and MITRE tactic sequence similarity.
* **3. Human-in-the-Loop SOAR Integration:** Add one-click approval buttons on the Defensive Playbook to push firewall blocklists directly to Palo Alto PAN-OS or isolate hosts via CrowdStrike Falcon API.
* **4. Local Air-Gapped Deployment with Fine-Tuned SecLLM:** Run quantized open-weight models (Llama-3.3-70B / Qwen-2.5-Coder) locally via Ollama/vLLM inside air-gapped classified SOC networks so sensitive incident logs never leave the enterprise perimeter.

6. SHORT, SWEET, ON-POINT 90-SECOND HACKATHON PITCH GUIDE

0:00 - 0:25 | Person 1: The Problem & Research Gap

"SOC analysts drown in fragmented alerts -- a phishing email in Proofpoint, a CVE alert on the firewall, an EDR detection on a laptop. Enterprise platforms like OpenCTI take weeks to deploy, while pasting logs into raw LLMs causes dangerous hallucinations. We built Securify: an AI Threat Intelligence Fusion Engine powered by an original Open-Weight Verification Harness."

0:25 - 0:55 | Person 3: The Live Demo & Graph Correlation

"When we ingest 6 separate incident reports, Person 1’s deterministic engine automatically refangs obfuscated indicators like hxxps:// and [.], extracts 17 unique IPs, domains, hashes, and CVEs with 100% precision, and discovers 10 cross-report pivot bridges -- proving these 6 isolated alerts are actually one coordinated multi-stage APT attack."

0:55 - 1:30 | Person 2: The Open-Weight Model Harness & Actionable Defense

"Instead of trusting a black-box LLM, our Model Harness feeds only verified IOCs into our Open-Weight Llama 3.3 & Backboard.io Memory Harness. It maps the campaign across 14 MITRE ATT&CK techniques, strictly separates confirmed Evidence Facts from AI Hypotheses, and runs an automated post-generation Hallucination Check that guarantees 0% invented indicators -- giving SOC teams an immediate, trusted playbook of firewall blocks, SIEM hunts, and CVE patches."